

Narsi Reddy Sanikommu

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EDUCATION

Master of Science - Mechanical Engineering

Aug 2024 - May 2026

JOHNS HOPKINS UNIVERSITY

Baltimore, MD

Bachelor of Technology - Mechanical Engineering

Jul 2016- May 2020

National Institute of Technology (NIT Trichy)

Tiruchirappalli, India

EXPERIENCE

Becton, Dickinson, And Company (BD), Bio-sciences

Jul 2025 - Jan 2026

Mechanical Engineering Co-op Intern

Sparks, MD, USA

- Designed and validated a shuttle fixture for the **BD COR GX shuttle**, characterized tube puncture force and clamp-pin material force profiles; developed a spring-based mechanism that replicates tube puncture-force behavior for **20k cycles**, significantly improving test efficiency versus the prior few hundred cycle tube-replacement-intensive setup.
- Conducted detailed characterization of the **BD COR MX liquid-waste sensor** by developing a 3D contour map of sensor activation liquid volume as a function of sensor Z-height, and X-offset from the collector; enabled prediction of sensor trigger volume with **95%** accuracy to inform physical design decisions.
- Performed feasibility study and design iteration of the **BD COR PX load cell** using SolidWorks, including leak testing and final design validation; the proposed design enables real-time measurement of sample, diluent, and reagent weights, reducing sample waste, improving error detection, improving liquid spill issues and enhancing troubleshooting capability.
- Executed root-cause analysis (**RCA**) on the **BD COR MX extractor** magnetic-strength characterization fixture; implemented minimal design changes and material changes to eliminate fixture warping by adding a snap-fit support feature that consolidates PCB and protective cover loading forces into the metal holder, while not triggering revalidation of test tool.

Soft, autonomous microneedle rolls for oral drug delivery, Dr Gracias Lab

Oct 2024 - Nov 2025

Research Assistant

Baltimore, MD

- Designed and fabricated soft, autonomous microneedle roll devices using **photolithography & soft lithography** for targeted drug delivery in the GI tract to treat autoimmune disorders like **Crohn's disease and ulcerative colitis**.
- Optimized fabrication of drug-embedded gelatin microneedles (0.3 mm diameter) using soft lithography with sequential heating and centrifugation cycles, reducing process time from **2 hours to 2 minutes** and improving mold-release yield from **40% to 80%**.

Becton, Dickinson, And Company (BD)

Jul 2020 - Aug 2023

Design Engineer 2, Design Engineer 1, GET

Bangalore, India

- Led multiple PE projects involving material change by updating **SolidWorks** drawings per **GD&T** standards, planned and executed DV activities, analyzed data with **Minitab**, DV Protocols & report release, DV Reviews, and ECRO process through **Teamcenter PLM** and **SAP**. Completed projects 2 weeks ahead of schedule, avoided back-orders and operation delays.
- Coordinated **Test Method updates** by designing a new test fixture and performed Measurement System Analysis (**MSA**) and Finite Element Analysis (**FEA**) simulations to replicate fixture engagement with components. Updated and leveraged existing MSA, reducing DV tests by **5x** and saving **2 weeks** of testing time.
- Updated and validated **112 BOMs in SAP**, coordinating with plant stakeholders to avoid duplicate work; detected **5 BOM errors** early, preventing **3 weeks** of potential rework.
- Led **RCA** of high-pressure leak-test failures in **BD Alaris pump sets**; implemented fixture and work-instruction updates that eliminated re-testing, preserved **100%** of DV test data, and avoided an estimated **\$20K** in retest costs and schedule delays.
- Innovatively leveraged existing BD PIVC platform components to create a new HDS product prototype in **1 week at a cost of \$100**; led internal VoCs and validation activities, proposing a commercialization strategy that projected **6 months reduction in NPd timelines** and substantial regulatory and testing cost savings by utilizing pre-existing FDA-cleared data.
- Submitted **68 Invention Disclosure records**, with **14 IDR** filed as **US patents**, and some are in pipeline.

NOTABLE PROJECTS

Segmented Fluidics Flow for Soft-body actuation, & Self folding shell microfluidics | Baltimore, MD Sep 2025 - Present

- Explored segmented fluidic-flow actuation in soft-bodied systems by developing a single-channel hydrogel (**PEGDA-AAm hydrogel**) "noodle" actuator; implemented segment-wise expansion control using timed segmented fluid delivery (water) and dehydration cycles (IPA) to achieve flexible robot motion without any external actuation mechanisms.
- Investigated a **multimodal, single-channel microfluidic platform** to route drugs and nutrients to multiple target sites; applied the approach to self-folding shell organoids and developed low-cost, reusable SLA-printed prototypes.

SKILLS

Programming Languages Python | C++ | C

Software Microsoft Projects | Minitab | Arduino IDE | MATLAB | Processing code | ROS2 | CorelDRAW | KiCAD | CAD modeling (SolidWorks, CATIA, ANSYS, Abaqus, Creo, AutoCAD, NX CAD, Onshape) | Teamcenter PLM | NumPy

Technical Skills GD&T | SAP Transactions | Printing (FDM, SLA, DLS, BMF) | RCA | CAPA | Product Innovation | Product development | Design Controls and Engineering changes | MSA | DFSS | DFMA | Photolithography | Soft lithography | Microfluidic device design | MEMS Device Fabrication | Transmission Electron Microscopy | Scanning Electron Microscopy

NOTABLE PUBLICATIONS

Stable flow regulator assembly: US20240175499A1 (Application granted)

Fluid connector system: US12186518B2 (Application granted)

Comprehensive list and details of the patents: ([Google Scholar](#))